$$C^{(21)}_{\phi\;\partial^2}{}^{(21)} \to$$

$$
\begin{aligned}
&\frac{1}{16\,\pi^2}\,\Bigg(\frac{1}{2}\,g_1{}^4\,m_1{}^2\,\tilde{\mu}^2\,\mathrm{LF}_{3,2,0}[m_1,\,\tilde{\mu}] - \frac{1}{2}\,g_1{}^4\,m_1{}^2\,\tilde{\mu}^2\,\mathrm{LF}_{3,3,-1}[m_1,\,\tilde{\mu}] - \frac{1}{2}\,g_1{}^4\,m_1{}^2\,\tilde{\mu}^2\,\mathrm{LF}_{4,2,-1}[m_1,\,\tilde{\mu}] + \\
&\quad \frac{3}{2}\,g_2{}^4\,m_2{}^2\,\tilde{\mu}^2\,\mathrm{LF}_{3,2,0}[m_2,\,\tilde{\mu}] - \frac{3}{2}\,g_2{}^4\,m_2{}^2\,\tilde{\mu}^2\,\mathrm{LF}_{3,3,-1}[m_2,\,\tilde{\mu}] - \frac{3}{2}\,g_2{}^4\,m_2{}^2\,\tilde{\mu}^2\,\mathrm{LF}_{4,2,-1}[m_2,\,\tilde{\mu}] - \\
&\quad \frac{1}{8}\,g_1{}^4\,m_1{}^2\,\tilde{\mu}^2\,\mathrm{LF}_{4,2,-1}[\tilde{\mu},\,m_1] + \frac{3}{8}\,g_2{}^4\,m_2{}^2\,\tilde{\mu}^2\,\mathrm{LF}_{4,2,-1}[\tilde{\mu},\,m_2] - \\
&\quad 2\,m_1\,m_2\,g_1{}^2\,g_2{}^2\,\tilde{\mu}^2\,\mathrm{LF}_{2,2,1,0}[m_2,\,\tilde{\mu},\,m_1] + 2\,m_1\,m_2\,g_1{}^2\,g_2{}^2\,\tilde{\mu}^2\,\mathrm{LF}_{3,2,1,-1}[m_2,\,\tilde{\mu},\,m_1] + \\
&\quad \frac{1}{2}\,m_1\,m_2\,g_1{}^2\,g_2{}^2\,\tilde{\mu}^2\,\mathrm{LF}_{4,1,1,-1}[\tilde{\mu},\,m_1,\,m_2] + 2\,m_1\,m_2\,g_1{}^2\,g_2{}^2\,\tilde{\mu}^2\,\mathrm{LF}_{3,2,1,-1}[\tilde{\mu},\,m_2,\,m_1] - \\
&\quad \frac{3}{2}\,\tilde{\mu}^2\,y_d{}^{pt}\,y_d{}^{sr}\,\overline{a_d}^{pr}\,\overline{a_d}^{st}\,\mathrm{LF}_{2,1,1,1,0}\left[m_{\tilde{d}}{}^r,\,m_{\tilde{d}}{}^t,\,m_{\tilde{q}}{}^p,\,m_{\tilde{q}}{}^s\right] + \\
&\quad \frac{3}{2}\,\tilde{\mu}^2\,y_d{}^{pt}\,y_d{}^{sr}\,\overline{a_d}^{pr}\,\overline{a_d}^{st}\,\mathrm{LF}_{3,1,1,1,-1}\left[m_{\tilde{d}}{}^r,\,m_{\tilde{d}}{}^t,\,m_{\tilde{q}}{}^p,\,m_{\tilde{q}}{}^s\right] + \\
&\quad \frac{3}{4}\,\tilde{\mu}^2\,y_d{}^{pt}\,y_d{}^{sr}\,\overline{a_d}^{pr}\,\overline{a_d}^{st}\,\mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{d}}{}^r,\,m_{\tilde{q}}{}^p,\,m_{\tilde{d}}{}^t,\,m_{\tilde{q}}{}^s\right] + \frac{3}{4}\,\tilde{\mu}^2\,y_d{}^{pt}\,y_d{}^{sr}\,\overline{a_d}^{pr}\,\overline{a_d}^{st} \\
&\qquad \mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{d}}{}^r,\,m_{\tilde{q}}{}^s,\,m_{\tilde{d}}{}^t,\,m_{\tilde{q}}{}^p\right] - \frac{1}{2}\,\tilde{\mu}^2\,y_e{}^{pt}\,y_e{}^{sr}\,\overline{a_e}^{pr}\,\overline{a_e}^{st}\,\mathrm{LF}_{2,1,1,1,0}\left[m_{\tilde{e}}{}^r,\,m_{\tilde{e}}{}^t,\,m_{\tilde{l}}{}^p,\,m_{\tilde{l}}{}^s\right] + \\
&\quad \frac{1}{2}\,\tilde{\mu}^2\,y_e{}^{pt}\,y_e{}^{sr}\,\overline{a_e}^{pr}\,\overline{a_e}^{st}\,\mathrm{LF}_{3,1,1,1,-1}\left[m_{\tilde{e}}{}^r,\,m_{\tilde{e}}{}^t,\,m_{\tilde{l}}{}^p,\,m_{\tilde{l}}{}^s\right] + \frac{1}{4}\,\tilde{\mu}^2\,y_e{}^{pt}\,y_e{}^{sr}\,\overline{a_e}^{pr}\,\overline{a_e}^{st} \\
&\qquad \mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{e}}{}^r,\,m_{\tilde{l}}{}^p,\,m_{\tilde{e}}{}^t,\,m_{\tilde{l}}{}^s\right] + \frac{1}{4}\,\tilde{\mu}^2\,y_e{}^{pt}\,y_e{}^{sr}\,\overline{a_e}^{pr}\,\overline{a_e}^{st}\,\mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{e}}{}^r,\,m_{\tilde{l}}{}^s,\,m_{\tilde{e}}{}^t,\,m_{\tilde{l}}{}^p\right] + \\
&\quad \frac{1}{8}\,\tilde{\mu}^2\,y_e{}^{pt}\,y_e{}^{sr}\,\overline{a_e}^{pr}\,\overline{a_e}^{st}\,\mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{l}}{}^p,\,m_{\tilde{l}}{}^s,\,m_{\tilde{e}}{}^r,\,m_{\tilde{e}}{}^t\right] + \frac{3}{8}\,\tilde{\mu}^2\,y_d{}^{pt}\,y_d{}^{sr}\,\overline{a_d}^{pr}\,\overline{a_d}^{st} \\
&\qquad \mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{q}}{}^p,\,m_{\tilde{q}}{}^s,\,m_{\tilde{d}}{}^r,\,m_{\tilde{d}}{}^t\right] - \frac{9}{4}\,\tilde{\mu}^2\,y_d{}^{sr}\,y_u{}^{pt}\,\overline{a_d}^{pr}\,\overline{a_u}^{st}\,\mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{q}}{}^p,\,m_{\tilde{q}}{}^s,\,m_{\tilde{d}}{}^r,\,m_{\tilde{u}}{}^t\right] + \\
&\quad \frac{3}{4}\,\tilde{\mu}^2\,y_u{}^{pt}\,y_u{}^{sr}\,\overline{a_u}^{pr}\,\overline{a_u}^{st}\,\mathrm{LF}_{2,1,1,1,0}\left[m_{\tilde{q}}{}^p,\,m_{\tilde{q}}{}^s,\,m_{\tilde{u}}{}^r,\,m_{\tilde{u}}{}^t\right] - \frac{3}{4}\,\tilde{\mu}^2\,y_u{}^{pt}\,y_u{}^{sr}\,\overline{a_u}^{pr}\,\overline{a_u}^{st} \\
&\qquad \mathrm{LF}_{3,1,1,1,-1}\left[m_{\tilde{q}}{}^p,\,m_{\tilde{q}}{}^s,\,m_{\tilde{u}}{}^r,\,m_{\tilde{u}}{}^t\right] - \frac{3}{8}\,\tilde{\mu}^2\,y_u{}^{pt}\,y_u{}^{sr}\,\overline{a_u}^{pr}\,\overline{a_u}^{st}\,\mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{q}}{}^p,\,m_{\tilde{u}}{}^r,\,m_{\tilde{q}}{}^s,\,m_{\tilde{u}}{}^t\right] - \\
&\quad \frac{3}{8}\,\tilde{\mu}^2\,y_u{}^{pt}\,y_u{}^{sr}\,\overline{a_u}^{pr}\,\overline{a_u}^{st}\,\mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{q}}{}^p,\,m_{\tilde{u}}{}^t,\,m_{\tilde{q}}{}^s,\,m_{\tilde{u}}{}^r\right] - \\
&\quad \frac{3}{4}\,\tilde{\mu}^2\,y_u{}^{pt}\,y_u{}^{sr}\,\overline{a_u}^{pr}\,\overline{a_u}^{st}\,\mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{u}}{}^r,\,m_{\tilde{u}}{}^t,\,m_{\tilde{q}}{}^p,\,m_{\tilde{q}}{}^s\right]\Bigg)
\end{aligned}
$$